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Wildfire Impacts on Baseflow Dynamics in Snow-Dominated Mountain Catchments

Date: 2026-06-17 **Review Type:** Narrative Review **PRISMA Compliance:** Partial (search and screening documented; not a formal systematic review/meta-analysis)

Abstract

Background: Wildfire activity and severity have increased markedly across the western United States and other snow-dominated mountain regions over the past four decades, coincident with warming and earlier snowmelt. These mountain catchments supply a disproportionate share of regional water resources, making understanding of wildfire’s effect on low-flow (baseflow) hydrology a pressing applied and scientific question.

Objectives: This review synthesizes current evidence on how wildfire alters baseflow dynamics, with particular attention to snow-dominated catchments, addressing: (1) what physical mechanisms link wildfire to baseflow change; (2) what observational and modeling evidence shows about

the direction, magnitude, and duration of these changes; (3) how snow-specific processes (canopy loss, albedo, energy balance) modulate the wildfire–baseflow relationship; and (4) what major uncertainties and research gaps remain.

Methods: Literature was identified in two rounds. Round 1 used Claude Code’s native Web-Search/WebFetch tools (substituting for the `parallel-cli` tool specified in the literature-review skill, which was not installed) and yielded 14 sources. Round 2 used the dedicated `paper-search` CLI (querying CrossRef, OpenAlex, and Semantic Scholar) to systematically extend coverage; ten additional topic-targeted queries surfaced 234 unique candidate titles, of which 34 met inclusion criteria after screening. In total, 48 primary and review sources published 2010–2026 were selected and citation-verified (DOI resolution + CrossRef metadata, via `verify_citations.py`); this remains a narrative synthesis, not an exhaustive systematic review.

Results: Evidence converges on a few robust patterns and several persistent uncertainties. Wildfire reliably increases total streamflow and runoff ratio for several years post-fire, driven primarily by sharply reduced evapotranspiration (ET) following canopy and root loss. Effects specifically on *baseflow* (as opposed to total flow or peak flow) are more variable: some studies report increased baseflow fraction and reduced recession rates (consistent with enhanced groundwater recharge from reduced ET), while others report decreased recharge (consistent with reduced infiltration from soil hydrophobicity or increased evaporative loss from exposed soil/snow). In snow-dominated catchments specifically, canopy loss and ash/char deposition advance snowmelt timing by 1–4 weeks and increase peak shortwave radiation absorption by up to 200%, but the net effect on baseflow depends on whether earlier, faster melt translates into greater infiltration/recharge or greater direct runoff and sublimation loss — a question the literature has not resolved.

Conclusions: Wildfire’s net effect on baseflow in snow-dominated catchments is mechanistically plausible in either direction and empirically site-dependent, contingent on burn severity, pre-fire vegetation density, soil texture, climate seasonality, and snow zone (early/middle/late). The strongest research gap is the scarcity of paired pre-fire/post-fire groundwater and baseflow-specific monitoring in snow-dominated systems, as most foundational evidence comes from rain-dominated California and Colorado Front Range watersheds.

Registration: Not registered (narrative review). **Keywords:** wildfire, baseflow, low flow, snowmelt, groundwater recharge, evapotranspiration, montane catchments, western United States, post-fire hydrology

1. Introduction

1.1 Background and Context

Snowmelt-driven streamflow and groundwater recharge from mountain watersheds in the western United States supply water to roughly 70 million people across ten states (search finding, Section 2). At the same time, annual forest area burned by wildfire in the western US increased by over 1,100% between 1984 and 2020 (Williams et al., 2022), and burned area in late-melting, high-elevation “late snow zones” has grown significantly across 70% of western ecoregions since 1984 (Kampf et al., 2022). These two trends are converging on the same landscapes: high-elevation, forested, snow-dominated catchments that historically burned infrequently are now burning more often and more severely.

Baseflow — the slowly draining groundwater and shallow subsurface contribution to streamflow during dry periods — is disproportionately important in snow-dominated systems because it sustains late-summer and fall flows long after the snowmelt pulse has receded. Disruption of baseflow has direct consequences for aquatic ecosystems, water supply reliability, and irrigation/municipal planning. Yet baseflow response to wildfire is mechanistically more ambiguous than total streamflow or peak-flow response, because it depends on the net balance of several competing processes (reduced transpiration vs. reduced infiltration vs. altered snowmelt timing) that can each push baseflow in either direction.

1.2 Scope and Objectives

This review focuses specifically on baseflow/low-flow dynamics, with snow-dominated mountain catchments as the primary geographic and process lens. Rain-dominated and Mediterranean-climate studies are included where they establish baseflow mechanisms (recession rate analysis, groundwater recharge pathways) that are directly transferable to snow systems, but peak-flow, flood, and water-quality (sediment/nutrient) literatures are deliberately excluded except where needed for context.

Primary Research Questions: 1. By what physical mechanisms does wildfire alter baseflow generation and recession behavior? 2. What is the empirical direction, magnitude, and duration of wildfire’s effect on baseflow/low flow, and how consistent is it across studies? 3. How do snow-specific processes (canopy interception loss, albedo reduction, advanced melt timing) modulate the wildfire–baseflow relationship in mountain catchments? 4. What are the major unresolved questions and monitoring gaps?

1.3 Significance

Western US water-supply planning, drought management, and post-fire watershed rehabilitation all implicitly assume some model of how fire changes low flows, yet the review and synthesis literature (Ebel et al., 2025) explicitly identifies the magnitude and duration of wildfire effects on groundwater recharge and baseflow-sourced low flows as “poorly understood.” This review consolidates the current state of evidence to clarify what is established, what is contested, and where snow-dominated systems specifically diverge from the better-studied rain-dominated case.

2. Methodology

2.1 Search Strategy

Tooling note: The literature-review skill’s documented primary search tool, `parallel-cli`, was not available in this environment (not installed, and installation/authentication was not pursued for this session). Claude Code’s native `WebSearch` and `WebFetch` tools were substituted, using equivalent academic-domain-targeted queries. This is a methodological deviation from the skill’s default workflow and is disclosed here per the skill’s transparency requirements.

Round 1 — WebSearch/WebFetch (9 queries, run 2026-06-17): 1. “wildfire impact baseflow dynamics snow-dominated mountain catchments hydrology” 2. “post-fire streamflow recession groundwater recharge Western US snowmelt watershed” 3. “wildfire effect snowpack accumulation melt timing forest canopy loss mountain hydrology” 4. “post-wildfire evapotranspiration changes water yield increase forested watershed” 5. “burn severity soil hydraulic properties infiltration

subsurface flow paths wildfire” 6. “paired catchment study wildfire low flow baseflow recession constant long-term” 7. “wildfire hydrology review synthesis streamflow generation mechanisms snow-dominated” 8. “climate change increasing wildfire frequency severity compound effects mountain water supply future projections” 9. “vegetation recovery post-fire transpiration recovery time streamflow return to pre-fire baseline”

Follow-up targeted searches and WebFetch extractions were used to confirm full author lists, years, journals, and DOIs for each candidate source.

Round 2 — paper-search CLI (10 queries against CrossRef, OpenAlex, and Semantic Scholar, run 2026-06-18; up to 12 results per source per query): wildfire baseflow recession streamflow western United States; wildfire impact groundwater recharge forested catchment; wildfire forest canopy loss snowpack accumulation melt timing; post-wildfire evapotranspiration water yield forested watershed; burn severity soil hydraulic properties infiltration; paired catchment wildfire low flow baseflow recession; wildfire hydrology review streamflow generation mechanisms; climate change wildfire compounding effects mountain water supply; post-fire vegetation recovery transpiration streamflow recovery; snow-dominated catchment wildfire baseflow dynamics. This round returned 234 unique candidate titles across the three sources, all of which were screened against the criteria in §2.2 (see `sources/search_log.md` for the full candidate list and disposition).

All DOIs across both rounds were independently verified and metadata-confirmed against CrossRef via `verify_citations.py` (§2.4).

2.2 Inclusion and Exclusion Criteria

Inclusion: - Peer-reviewed journal articles, USGS Science publications, or AGU/PNAS articles - Direct treatment of post-fire streamflow, baseflow, low flow, groundwater recharge, snowpack, or evapotranspiration in forested/montane catchments - English language - No hard date cutoff, but priority given to 2015–2026 with select foundational mechanistic studies included

Exclusion: - Studies focused exclusively on peak flow/flood response, debris flow, or water quality (sediment, nutrients, contaminants) without a baseflow/low-flow component - Non-forested or urban wildland-urban-interface fire studies without catchment-hydrology framing - Preprints and non-peer-reviewed sources (none were ultimately needed given search yield) - Paywalled sources for which neither full text nor a verifiable abstract/findings summary could be obtained

2.3 Study Selection

Single-reviewer narrative screening (title → abstract → available full text), conducted in two rounds. Round 1 retained 14 sources after screening ~30 candidate titles surfaced across nine WebSearch queries. Round 2 screened 234 unique candidate titles/DOIs returned by the **paper-search CLI** (CrossRef + OpenAlex + Semantic Scholar) across ten topic-targeted queries; 34 met inclusion criteria and were added, none of which duplicated a Round 1 source. Excluded in Round 2: studies lacking any wildfire/fire treatment (general snow, climate, or ecology studies — the majority of candidates), preprints and conference abstracts (e.g., SSRN, GSA abstracts, ESSOAr), book chapters, water-quality/contaminant-only or debris-flow-only studies without a baseflow/streamflow component, and off-topic climate/ecology/agriculture hits unrelated to forested-catchment hydrology. Combined, 48 sources were retained.

2.4 Data Extraction

For Round 1 sources, author list, year, journal, DOI, study location/design, and quantitative findings were extracted via WebFetch where full text was accessible (open-access USGS, PNAS via PMC mirror, IOPscience, AGU) or via search-result synthesis where paywalled (ScienceDirect, Wiley — abstract-level information only, confirmed against at least two independent search results before inclusion). For Round 2 sources, abstracts and metadata (authors, year, journal, volume, pages, DOI) were retrieved directly from the `paper-search` CLI’s CrossRef/OpenAlex/Semantic Scholar results, and every DOI (Round 1 and Round 2 combined) was independently re-verified via `verify_citations.py`, which resolves each DOI and cross-checks metadata against the CrossRef API (48/48 verified, 0 failed — see `wildfire_baseflow_review_citation_report.json`).

3. Results

3.1 Study Selection Summary

48 sources retained: 6 review/synthesis articles (Ebel et al., 2025; Goeking & Tarboton, 2020; Guzmán-Rojo et al., 2024; Robinne et al., 2021; Walvoord et al., 2025; Zacharakis & Tsihrintzis, 2026) and 42 primary observational or modeling studies, spanning streamflow/water-yield response, soil-hydraulic/infiltration response, snowpack/energy-balance response, and groundwater/baseflow-recession response. Years range 2010–2026 (median ~2020). Geographic coverage is dominated by the western US (the large majority of sources), with additional forested-catchment case studies from Italy, Iberia (Spain and Portugal), and Australia, plus several reviews of global or cross-regional scope. All sources are peer-reviewed journal articles; all 48 DOIs were independently verified (§2.4).

3.2 Thematic Synthesis

3.2.1 Theme: Evapotranspiration reduction is the dominant, most consistent mechanism Across nearly every study reviewed, the most robust and consistent finding is that canopy and root loss from wildfire sharply reduces evapotranspiration, and this ET reduction is the primary driver of increased total water yield. Blount et al. (2020) documented a ~250 mm (46%) annual ET decrease in a western US forested catchment following wildfire, accompanied by a 136 mm increase in annual streamflow (a 21% increase in runoff ratio, 140% increase in water yield), with no trend back toward pre-fire values over 10 years of post-fire record. Williams et al. (2022), analyzing 72 burned vs. 107 unburned control basins across the western US (1984–2019), found that streamflow in basins with >20% forest area burned was elevated by an average of ~30% (median 20%) for roughly 6 water years post-fire, with effects detectable in all four seasons and the largest absolute contribution to the annual increase coming from spring. Saxe et al. (2018), examining 82 western US watersheds, similarly found low flows, high flows, and peak flows all increase in the first ~2 years post-fire before declining, though with strong regional heterogeneity — coastal California watersheds showed the strongest response while eastern California, western Nevada, and Oregon sites showed minimal or even negative responses. Bart (2016)’s regional paired-watershed analysis of 12 central/southern California catchments quantifies this further: first-year postfire streamflow increase scales with both percent area burned and antecedent wetness, averaging 134% (range 82–200%) at 100% area burned under average wetness, with the smallest response in dry years and the largest in wet years — directly implicating *interaction* between fire and climate variability rather than fire alone as the proximate driver of yield magnitude. Hallema et al. (2018), synthesizing 168 US fire-climate-flow records, found this scales unevenly across regions: river flow changed

significantly in only 32 of 168 locations where >19% of basin area burned, with the strongest enhancement in the semi-arid Lower Colorado region and no significant change from prescribed burns in the humid subtropical Southeast — underscoring that “ET reduction increases yield” is climate- and fire-type-contingent, not universal. At the continental scale, Wine, Makhnin, et al. (2018)’s Fuh’s-equation analysis attributes up to 20% of streamflow to wildfire effects in the most heavily burned parts of the western US (1986–2015), a magnitude that can rival projected climate-change streamflow reductions. Outside the western US, Khaledi et al. (2022) and Nolan et al. (2015) document comparable post-fire ET decline and streamflow/water-yield increases in Australian temperate and resprouting eucalypt forests respectively, and Poon & Kinoshita (2018) report consistent post-fire ET decline in semi-arid landscapes — suggesting the ET-reduction mechanism generalizes across forest types and climates, even where the downstream magnitude does not.

This consistent water-yield increase, however, conflates baseflow with total flow, and the ET-reduction narrative itself has an important caveat. Goeking & Tarboton (2020)’s synthesis of 78 disturbance-streamflow studies (standing-replacing disturbances including severe wildfire, and non-standing-replacing disturbances such as low-severity fire, drought, and insects) found that post-disturbance ET sometimes *increases* rather than decreases — particularly after non-stand-replacing disturbance — due to higher subcanopy radiation reaching surviving understory and rapid post-disturbance regrowth transpiration. This means the streamflow response is not a simple function of canopy loss alone but depends on vegetation structure, climate, and topography interacting with burn severity, and several of the same studies caution that the *partitioning* of any extra water between direct runoff, soil storage, and groundwater recharge is the open question relevant to baseflow specifically (Ebel et al., 2025).

3.2.2 Theme: Baseflow and recession-rate responses are directionally inconsistent across sites Evidence on baseflow specifically (rather than total or peak flow) is markedly less consistent than the ET/water-yield literature. Bart & Tague (2017), analyzing burned and unburned California catchments, found a 52.5% reduction in baseflow recession rates following wildfire — i.e., baseflow declined *more slowly* post-fire, which the authors attributed to diminished transpiration outweighing any hydrophobicity-driven infiltration loss, since reduced water uptake by surviving and burned vegetation left more water available to sustain groundwater discharge during the dry season. This is consistent with a recharge-enhancing interpretation of fire effects. Beyene et al. (2021)’s quantile-based statistical framework, applied to 44 western US streams, supports this directionally: large, high-severity burns elevated low-flow quantiles for at least five post-fire years, with the increase in low flow (up to several hundred percent at the lowest quantiles in some streams) often exceeding the proportional increase at high-flow quantiles — though the magnitude varied strongly by region (most pronounced in the Pacific Northwest and California) and was confounded by post-fire weather, underscoring that apparent “baseflow enhancement” signals are sensitive to the statistical method and post-fire climate window used to detect them.

In contrast, the Mt. Pisano (Tuscany, Italy) paired-catchment study (Nigro et al., 2026) — using stream discharge combined with chemical and stable-isotope tracers — found the opposite signature in a Mediterranean mountain setting: burned catchments showed increased stormflow, a stronger rock–water interaction signal in baseflow, and a stronger evaporative signal in baseflow, all interpreted as evidence of *reduced* groundwater recharge after fire. Newcomer et al. (2023), studying ten wildfires across the drought-prone, Mediterranean-climate Russian River Watershed in coastal northern California, found a third possible outcome — essentially *no* detectable hydrologic response — even where burned-area ratios exceeded previously reported disturbance thresholds

(>45% in places), attributing the null result to a combination of regional fire adaptation and concurrent drought suppressing any wildfire-attributable signal; they propose that hydrologic response to fire only manifests above some watershed-specific burn-severity/burn-area threshold, which complicates cross-site comparison of baseflow studies that do not report or control for burn extent. Two Mediterranean Portugal headwater-catchment studies reinforce this climate-modulation point from outside the western US: Nunes et al. (2020), examining a humid Mediterranean headwater catchment before and after wildfire plus post-fire land management, and Wu et al. (2021), comparing the same catchment’s hydrological and erosion responses to wildfire against its responses to meteorological variability, both found that — despite moderate-to-high burn severity across much of the catchment — meteorological variability exerted stronger control over post-fire hydrological response than the fire itself, with only high-return-period (~2-year) rainfall events producing a detectable fire-attributable signal during the vegetation-recovery window. Taken together with Newcomer et al. (2023), this is now a recurring pattern across three independent, climatically distinct catchments: wildfire’s hydrologic signature can be readily masked by concurrent or antecedent weather conditions, a confound that any baseflow-attribution study design needs to explicitly control for.

Walvoord et al. (2025)’s WIREs Water review is the most direct attempt yet to systematize this inconsistency: synthesizing post-wildfire groundwater studies, it organizes the “disparate” outcomes (substantial increases, notable decreases, and negligible/short-lived effects in recharge and baseflow) into five interacting control categories — climate, vegetation, hydrogeology, fire characteristics, and the cryosphere — and proposes that, as a general pattern, groundwater response tends to be *positive* where water-input and evaporative-demand cycles are decoupled (e.g., wet-season recharge after a dry, high-ET growing season), and *negative* under low fire severity or in vegetation types that recover quickly. Critically, Walvoord et al. (2025) explicitly single out the cryosphere category as the least developed: “a key area for further research is understanding how wildfire effects on snow dynamics and other cryospheric processes translate to changes in groundwater” — directly corroborating, from an independent 2025 review, the same snow–baseflow linkage gap that Ebel et al. (2025) identify (§3.2.3, §3.3). Ebel et al. (2025)’s own framing of the underlying tension remains apt: two competing conceptual models exist — (1) reduced infiltration (from soil sealing/hydrophobicity) decreasing recharge, versus (2) reduced transpiration combined with greater throughfall increasing recharge — and the review also notes a critical methodological problem: most burned catchments lack pre-fire groundwater monitoring, and compensatory water uptake by surviving unburned vegetation can mask or dampen the recharge signal that would otherwise be detectable in baseflow records.

The soil-hydraulic-properties literature helps explain why both outcomes (and a null outcome) are physically plausible, and recent additions sharpen the burn-severity dependence considerably. Moody et al. (2016)’s foundational synthesis established that post-fire changes in saturated hydraulic conductivity and sorptivity are nonlinear in burn severity, with a threshold (dNBR 124–420) below which properties are statistically unchanged, and above which conductivity decreases exponentially. Ebel et al. (2018), resampling the 2011 Las Conchas Fire (New Mexico) site four years post-fire, found that several physical thresholds (bulk density, soil-water content, gravel/fines fraction) remained detectable at dNBR 293–618, but found *no* persistent threshold or effect on field-saturated hydraulic conductivity or sorptivity at that site by year four — implying the infiltration-capacity signal, while real shortly after fire (Ebel et al. (2012) found that at Colorado’s 2010 Fourmile Canyon Fire, an ash layer alone stored 97–99% of rainfall and controlled whether runoff was generated by saturation-excess or infiltration-excess mechanisms in the weeks after burning), can fade within a few years even where other soil-physical changes persist. Moody et al. (2019) further show that infiltration variability itself is partly an artifact of measurement (ash/sediment

layering adds substantial extra variance beyond burn severity alone), while Ebel et al. (2022), revisiting Colorado’s 2013 Black Forest Fire, and McGuire & Youberg (2019), studying twice-burned soils in Arizona’s Pinaleno Mountains, both show that burn-severity-conductivity relationships are history-dependent — soils burned a second time at even low severity can retain reduced infiltration capacity from the prior fire, indicating the effective recovery clock can reset incompletely after repeat burns. Comparable burn-severity/hydraulic-conductivity relationships are documented outside the western US conifer context: Plaza-Álvarez et al. (2019) and Fernández et al. (2021) in Mediterranean/Spanish pine stands, and Frimpong et al. (2026) via laboratory rainfall-simulation experiments showing that hydraulic-conductivity reduction in high-burn-severity soils does not appear immediately but only *after* the first post-fire rainfall event — a timing nuance largely absent from the original 2016 synthesis. Taken together, this body of work means the infiltration-reduction pathway to lower recharge is burn-severity- and time-since-fire-dependent and will not manifest uniformly even within a single fire perimeter — consistent with the MDPI review’s (Guzmán-Rojo et al., 2024) conclusion that quantifying and generalizing fire-induced recharge change remains methodologically immature, particularly regarding how to integrate fire-altered precipitation, wind, and land-surface temperature into recharge projection models.

Two further studies add nuance specifically relevant to subsurface flow pathways. Maina & Siirila-Woodburn (2020), using a high-resolution integrated hydrologic model of California’s Cosumnes River (Sierra Nevada), found a genuinely counterintuitive feedback directly bridging the snow and groundwater themes: postfire simulations showed greater winter snowpack accumulation in some regions (less canopy interception) translating into greater summer runoff *and* greater groundwater storage, even as catchment-average ET decreased — one of the only studies in this review to mechanistically connect altered snow accumulation to a groundwater-storage outcome within the same modeling framework. Atwood et al. (2023), combining electrical resistivity imaging and stable-isotope analysis in three San Gabriel Mountains (California) catchments burned by the 2020 Bobcat Fire, found that both surface runoff *and* subsurface infiltration into weathered bedrock increased in tandem after fire — challenging the simpler conceptual model in which post-fire runoff increases necessarily imply reduced recharge, and suggesting more surface–subsurface exchange occurs during storms than previously assumed.

3.2.3 Theme: Snow-specific mechanisms add a layer of complexity unique to mountain catchments Snow-dominated catchments introduce mechanisms absent from rain-dominated systems, and the literature on these mechanisms is comparatively mature even though their downstream effect on baseflow is not yet resolved. Kampf et al. (2022), using field measurements at Colorado’s Cameron Peak Fire plus a broader western US analysis, found burned sites had earlier and lower-magnitude peak snow water equivalent (SWE) and an estimated 18–24 day advance in snow-free dates; canopy loss caused a 60% increase in solar radiation reaching the snow surface and a 200% increase in net shortwave radiation absorbed by the snowpack, with incoming shortwave dominating over albedo change as the controlling energy term. Giovando & Niemann (2022), using 45 burned SNOTEL sites against unburned controls, corroborate accelerated snowmelt phenology attributable specifically to fire rather than concurrent climate warming. Maxwell et al. (2019) add a topographic dimension: canopy loss can paradoxically *increase* snow accumulation on the ground (less interception by scorched, needle-less trees) even as it accelerates melt via radiation exposure, with the net retention/melt-timing outcome depending strongly on slope aspect — south-facing burned slopes lose snowpack fastest. Gleason et al. (2019), measuring snow at seven burned western US forest sites (2002–2016), quantify the radiative mechanism at decadal scale: black carbon and charred woody debris darken the snowpack and depress albedo for up to 15 winters post-fire,

driving a 372–443% increase in solar energy absorbed by snowpacks over the past two decades and causing earlier snow disappearance in >11% of the western seasonal snow zone by 2018 — a persistence considerably longer than the multi-year soil-hydraulic effects discussed above. Hatchett et al. (2023) show this albedo effect interacts strongly with weather sequencing: at two California burned sites, accumulation-season albedo declined 25–71% relative to unburned conditions, and this reduced albedo (combined with canopy loss) drove substantial midwinter snowpack melt during a multi-week dry spell in 2022, whereas a meteorologically similar dry spell in 2013 (before fire) produced minimal melt — demonstrating that post-fire snow vulnerability is conditional on compounding with specific weather sequences, not a fixed seasonal offset.

The unresolved question, as Ebel et al. (2025) and, independently, Walvoord et al. (2025) explicitly state, is whether this earlier/faster snowmelt translates into greater streamflow and groundwater recharge or not: “tradeoffs between snow sublimation and the rate and timing of snowpack ablation can make peak SWE an inaccurate proxy for snowmelt volume” (Ebel et al., 2025). Earlier melt could enhance recharge by extending the window of low-evapotranspiration-demand infiltration in early spring, or it could reduce recharge if faster melt overwhelms infiltration capacity and is lost to direct runoff, or if it shifts melt earlier into a period when frozen/low-permeability soils limit infiltration. Two studies in this expanded review come closest to closing this loop, though neither does so completely. Seibert et al. (2010), modeling the Entiat Experimental Forest (Washington State) pre- and post-fire, found that post-fire conceptual-model parameters indicated deeper snowpacks with earlier, more rapid snowmelt, and that observed post-fire peak flows averaged 120% higher than pre-fire-calibrated model predictions — directly linking the snow-timing mechanism to a streamflow (though not baseflow-specific) outcome at the same site later revisited by Niemeyer et al. (2020) (§3.2.4). Maina & Siirila-Woodburn (2020) (§3.2.2) provide the clearest groundwater-specific link, showing in the Cosumnes River model that postfire winter snowpack increases in some sub-regions translated into greater summer runoff and groundwater storage — but this remains a single modeling study, and Ebel et al. (2025)’s call for paired snow-timing/baseflow-recharge field observations in the same snow-dominated catchment remains, on the evidence assembled here, still largely unanswered.

3.2.4 Theme: Recovery timescales are long, variable, and ecoregion-dependent — with implications for cumulative/compound fire risk

Wine, Cadol, et al. (2018), modeling 39 gauged western US basins across seven ecoregions, attributed 2–14% of long-term annual streamflow to wildfire effects, with vegetation recovery timescales ranging from 1.8 years (tropical/subtropical steppe) to 12.5 years (marine-regime mountains) — directly relevant to snow-dominated systems, which fall toward the longer end of this range. Zacharakis & Tsihrintzis (2026)’s broader review concurs that abiotic (soil/hydrologic) recovery typically occurs within about a decade under a “typical” fire regime and temperate climate, while vegetation structural recovery lags 15–25 years. Williams et al. (2022) found the postfire streamflow enhancement itself persists for an average of 6 water years before declining toward pre-fire-adjusted expectations. The longest empirical record in this review, Niemeyer et al. (2020)’s 35–41-year follow-up at the Entiat Experimental Forest (the same site modeled by Seibert et al. (2010)), shows recovery is not just slow but management-contingent: two salvage-logged-and-seeded catchments returned to pre-fire discharge, peak-flow, low-flow, and ET levels by 35–41 years post-fire, while the third catchment — burned but left unmanaged — remained elevated in annual discharge and runoff ratio (with AET still depressed) at the same 35–41-year mark, attributed to persistently delayed vegetation recovery.

Post-fire forest-management interventions add a further complication to recovery trajectories and to

the magnitude of the wildfire signal itself. Saksa et al. (2020), modeling two Sierra Nevada fireheds, found wildfire alone increased runoff 55–67% (with vegetation loss of 38–50%), versus a smaller 3–12% runoff increase from mechanical fuels treatments alone (8% vegetation loss) — fuels treatments produce a qualitatively similar but much smaller hydrologic signal than wildfire itself, which matters for interpreting “treated” catchments as a counterfactual for “unburned.” Srivastava et al. (2018), simulating a Pacific Northwest watershed, similarly found fuel treatments (thinning plus prescribed burn) increased water yield only 0.7–5.6% at the watershed scale despite much larger hillslope-scale changes in surface runoff, lateral flow, and ET — implying treatment effects can be diluted at the scales most relevant to baseflow monitoring. Furniss et al. (2025)’s coupled LANDIS-II/DHSVM simulation of a western US mountain landscape over the 21st century frames the larger stakes: wildfire and forest-thinning treatments can partially offset climate-driven snowpack and spring-flow declines, but neither management practice nor wildfire itself can fully prevent a projected regime shift from spring snowmelt- to autumn rain-dominated streamflow as winter temperatures warm — meaning the recovery-timescale question this review addresses is itself nested inside a longer-term, non-stationary climate trend in snow-dominated systems. A complementary, longer-timescale management contrast comes from Boisramé et al. (2017), who studied multiple decades of *managed* wildfire (a let-burn policy permitting repeated, largely unsuppressed fire) in the Illilouette Creek basin, Yosemite National Park: relative to a fire-suppressed comparison basin, the repeatedly burned basin developed greater landscape-scale forest heterogeneity, which was associated with increased soil moisture and more persistent late-season stream flow — suggesting that, at a multi-decadal, basin-wide scale, a managed-fire regime can produce more favorable water-availability outcomes than either single-fire-event suppression-and-recovery cycles or active post-fire salvage management (cf. Niemeyer et al. (2020), above).

These long and heterogeneous recovery timescales matter for baseflow specifically because they set the window during which altered ET/infiltration/recharge processes operate, and because increasing fire frequency under climate change (Williams et al. (2022) project 85–750% more frequent extreme aridity conditions by 2021–2050) raises the likelihood of catchments reburning before baseflow-relevant vegetation and soil properties have recovered — a risk directly evidenced by McGuire & Youberg (2019)’s twice-burned Arizona soils (§3.2.2), which is the clearest empirical case of incomplete recovery before reburn in this review.

3.2.5 Theme: Wildfire is an increasingly material — and increasingly recognized — water-supply risk A cluster of perspective and synthesis pieces, mostly outside the primary streamflow/baseflow measurement literature, frames why the mechanistic questions above carry practical urgency. Bladon et al. (2014) and Robinne et al. (2021) — the latter explicitly titled “Scientists’ warning on extreme wildfire risks to water supply” and citing the 2019–2020 Australian and Pantanal fire seasons as case-in-point — both argue that wildfire’s effects on downstream water treatment costs, source-water contamination, and supply reliability are an underappreciated, escalating risk requiring proactive (rather than reactive) governance and combined “green” (watershed restoration) and “grey” (treatment infrastructure) investment. Barnard et al. (2023) sharpen this to the mountain-source-water context central to this review, arguing that wildfire and climate change jointly amplify existing knowledge gaps in how mountain source-water systems feed downstream agricultural water supply in the western US — a framing that places the baseflow-mechanism uncertainty documented in §3.2.2–3.2.3 directly upstream of unresolved water-management and agricultural-planning questions. None of these pieces resolve the baseflow-direction question; their contribution is to establish that the stakes of resolving it are rising in step with wildfire trends documented elsewhere in this review (Kampf et al., 2022; Williams et al., 2022). Gould et al. (2016),

modeling a mountain watershed under combined climate-change and extreme-wildfire scenarios, provide one of the few studies in this review to explicitly quantify this compounding: projected warming and increased wildfire severity amplified runoff and erosion beyond what either driver produced alone — a reminder that the baseflow-relevant mechanisms reviewed in §3.2.1–3.2.4 are not occurring against a stationary climate baseline, and that future hydrologic risk assessments need to account for fire and climate change as interacting rather than additive drivers.

3.3 Knowledge Gaps

Mechanistic/Process gaps: No reviewed study fully closes the loop from snowmelt-timing change to a measured baseflow-recharge outcome within the same catchment, though Maina & Siirila-Woodburn (2020) and Seibert et al. (2010) come closest by linking altered snow accumulation/melt timing to streamflow or groundwater-storage outcomes in process models at single sites. The ET-reduction-vs-infiltration-reduction tension for recharge remains conceptually unresolved at a general level, though Walvoord et al. (2025)’s five-category framework (climate, vegetation, hydrogeology, fire characteristics, cryosphere) now provides a structured set of testable hypotheses for resolving it — the framework itself flags the cryosphere category as its least-developed component.

Monitoring/Data gaps: Pre-fire groundwater and baseflow-specific monitoring is rare, especially in snow-dominated catchments; most baseflow-specific evidence (Bart & Tague, 2017) comes from rain-dominated California systems, limiting direct transferability to snow-dominated mountain settings. Long-term (multi-decade) post-fire hydrologic records remain rare — Niemeyer et al. (2020)’s 35–41-year Entiat record is the clear exception in this review rather than the norm.

Methodological gaps: Burn-severity thresholds for infiltration change (Moody et al., 2016) are increasingly refined by time-since-fire (Ebel et al., 2018; Frimpong et al., 2026) and repeat-burn history (McGuire & Youberg, 2019), but are not yet incorporated into most catchment-scale recharge/baseflow studies, which tend to treat “burned” as a binary condition. Newcomer et al. (2023)’s finding that hydrologic response may not manifest below a watershed-specific burn-area threshold further complicates cross-study comparison where burn extent is not consistently reported.

Compound-disturbance gaps: Reburn risk under shortening fire-return intervals, and its effect on baseflow recovery trajectories, is identified as a forward-looking concern (Williams et al., 2022) but not yet empirically characterized.

4. Discussion

4.1 Main Findings

1. Wildfire consistently and substantially increases total water yield via ET reduction; this is the best-established finding in the literature, though Goeking & Tarboton (2020) show ET can also increase post-disturbance under some vegetation/severity combinations, and the magnitude is strongly climate- and region-dependent (Bart, 2016; Hallema et al., 2018).
2. The baseflow-specific signal is directionally inconsistent across studies and sites — enhanced recharge/slower recession (Bart & Tague, 2017; Beyene et al., 2021), reduced recharge (Nigro et al., 2026), and null/undetectable response under drought (Newcomer et al., 2023) are all documented, with burn severity, time since fire, repeat-burn history, vegetation recovery

- rate, and soil hydraulic thresholds as plausible moderators. Walvoord et al. (2025)’s five-category synthesis (climate, vegetation, hydrogeology, fire characteristics, cryosphere) is the most systematic attempt yet to organize this inconsistency, though it does not yet resolve it.
3. Snow-dominated catchments add radiation-balance and snowmelt-timing mechanisms (Giovando & Niemann, 2022; Gleason et al., 2019; Hatchett et al., 2023; Kampf et al., 2022; Maxwell et al., 2019) that are well characterized in their own right; two studies (Maina & Siirila-Woodburn, 2020; Seibert et al., 2010) provide single-site modeling evidence linking altered snow timing to a streamflow or groundwater-storage outcome, but a direct, field-paired snow-timing/baseflow-recharge link remains absent from the broader literature.
 4. Recovery timescales are long (years to over a decade, and up to 35–41 years for full hydrologic recovery at the unmanaged Entiat site; Niemeyer et al. (2020)) and are management-contingent as well as ecoregion-dependent, raising compound-disturbance concerns as fire frequency increases — concerns made more concrete by McGuire & Youberg (2019)’s direct evidence of incomplete infiltration-capacity recovery before reburn.
 5. The broader water-supply-risk literature (Barnard et al., 2023; Bladon et al., 2014; Robinne et al., 2021) frames these mechanistic uncertainties as carrying escalating real-world stakes for downstream water treatment, supply reliability, and agricultural water management, independent of which baseflow direction ultimately dominates at a given site.

4.2 Interpretation and Implications

The field has good answers for “does wildfire increase streamflow” and decent mechanistic answers for “why” (ET reduction, canopy loss, soil hydraulic change), but a genuinely open answer for “does wildfire increase or decrease the groundwater-sustained low-flow baseline, especially after the snow has melted” — which is often the more policy-relevant question for late-summer water supply and ecological flow requirements. The expanded evidence base strengthens rather than resolves this picture: Walvoord et al. (2025)’s five-category framework gives the field a structured set of testable moderators (climate decoupling, vegetation recovery rate, hydrogeologic setting, fire characteristics, cryospheric processes) rather than a single answer, and the soil-hydraulics literature (Ebel et al., 2018; Ebel et al., 2022; Frimpong et al., 2026; McGuire & Youberg, 2019) makes clear that burn severity alone is an incomplete predictor without time-since-fire and burn-history context. For research and proposal design (e.g., studies targeting post-wildfire baseflow in snow-dominated WUS watersheds), this gap is the clearest opportunity: a study design pairing snow-timing observations, soil-hydraulic burn-severity/time-since-fire classification, and multi-year baseflow recession analysis in the same set of catchments would directly address the central unresolved tension identified by Ebel et al. (2025) and operationalize the cryosphere category that Walvoord et al. (2025) flag as their framework’s weakest link.

4.3 Strengths and Limitations

Strengths: Coverage spans the full causal chain from energy balance (snow) through soil hydraulics through catchment-scale streamflow response, anchored by two independent, comprehensive 2025 reviews/syntheses (Ebel et al.; Walvoord et al.) that both frame the open questions similarly despite being authored by different teams. The evidence base now includes geographically diverse confirmatory and contrasting case studies (Mediterranean Europe, Australia) and several multi-decade observational records (Niemeyer et al., 2020; Seibert et al., 2010), strengthening confidence that the directional inconsistency documented for baseflow is a real feature of the system rather than an artifact of a narrow evidence base.

Limitations: - Single-reviewer narrative screening, not independently duplicated, across both search rounds. - Round 1 used native WebSearch/WebFetch tooling (substituted for the skill’s specified `parallel-cli`); Round 2 used the dedicated `paper-search` CLI (CrossRef, OpenAlex, Semantic Scholar). The two tools may have different recall/precision characteristics, and some ScienceDirect/Wiley sources were only accessible at abstract/snippet level (paywalled), so nuanced methodological details of those studies could not be independently verified beyond what search engines or abstracts surfaced. - Geographic coverage remains dominated by the western US; non-US case studies (Italy, Spain/Portugal, Australia) and cross-regional reviews (Goeking & Tarboton, 2020; Hallema et al., 2018) broaden but do not eliminate this skew, and direct snow-dominated-catchment baseflow evidence outside North America is still sparse. - This is a narrative synthesis of 48 sources, not an exhaustive systematic review; additional relevant literature (e.g., on isotope-based baseflow source attribution, or specific WUS long-term experimental watersheds like H.J. Andrews) likely exists but was not exhaustively pursued given the scope of this session.

4.4 Future Research

Priority questions: 1. Does advanced snowmelt timing post-fire increase or decrease net ground-water recharge, holding burn severity constant? — Requires paired snow-timing/recharge instrumentation in the same catchments; directly addresses the central gap identified by Ebel et al. (2025) and the cryosphere category flagged as underdeveloped by Walvoord et al. (2025). Maina & Siirila-Woodburn (2020)’s Cosumnes River modeling framework and Seibert et al. (2010)’s Entiat snow-hydrology model offer plausible methodological templates for a field-validated follow-up (for a review of physically based distributed post-fire hydrologic models from which such templates can be drawn, see Ebel et al. (2023)). 2. Can burn-severity *and time-since-fire* infiltration thresholds (Ebel et al., 2018; Ebel et al., 2022; Frimpong et al., 2026; Moody et al., 2016) jointly predict catchment-scale baseflow recession-rate change? — Requires linking plot-scale, repeat-sampled soil hydraulic measurements to catchment-integrated recession analysis. 3. How does baseflow recovery trajectory change under shortened fire-return intervals (reburn before full recovery)? — McGuire & Youberg (2019) provide a soil-hydraulic-properties precedent for incomplete recovery before reburn; extending this to catchment-scale baseflow recession would require either long-term monitoring in repeatedly burned basins or process-based modeling calibrated against the recovery timescales reported by Niemeyer et al. (2020), Wine, Cadol, et al. (2018), and Zacharakis & Tsihrintzis (2026). 4. Can Walvoord et al. (2025)’s five-category framework be operationalized as a predictive (rather than retrospective/organizing) tool — i.e., can climate, vegetation, hydrogeology, fire-characteristic, and cryosphere variables be combined into a regional model that forecasts baseflow direction at unstudied sites?

5. Conclusions

1. Wildfire reliably increases total post-fire streamflow in forested catchments, primarily via evapotranspiration reduction — this is well established across rain- and snow-dominated systems alike, though the magnitude is strongly climate-, region-, and fire-type-dependent, and ET occasionally increases rather than decreases post-disturbance (Goeking & Tarboton, 2020).
2. Wildfire’s effect on baseflow specifically is not unidirectional in the existing literature: enhanced recharge/slower recession, reduced recharge, and null/undetectable responses are all documented (§3.2.2), and the net direction appears to depend on the balance between reduced

transpiration (favoring recharge), reduced infiltration capacity from burn severity (favoring reduced recharge), and confounding factors including time since fire, repeat-burn history, and concurrent drought. Walvoord et al. (2025)’s five-category synthesis framework is the most systematic attempt to date to organize, though not yet resolve, this inconsistency.

3. Snow-dominated catchments introduce well-characterized but mechanistically incompletely-resolved snowmelt-timing and energy-balance effects; two single-site modeling studies (Maina & Siirila-Woodburn, 2020; Seibert et al., 2010) link altered snow dynamics to streamflow or groundwater-storage outcomes, but downstream consequences for baseflow specifically remain a clear research gap, due primarily to a scarcity of paired pre-/post-fire groundwater and baseflow monitoring in these systems.
4. Recovery is slow and contingent on post-fire forest management: the longest available record (35–41 years, Niemeyer et al. (2020)) shows full hydrologic recovery in actively managed (salvage-logged, seeded) catchments but persistent elevation in an unmanaged comparison catchment at the same site, indicating that “recovery timescale” is not a fixed property of climate or ecoregion alone.
5. The escalating water-supply-risk literature (Barnard et al., 2023; Bladon et al., 2014; Robinne et al., 2021) underscores that resolving the baseflow-direction question has rising practical stakes for downstream water treatment, supply reliability, and agricultural water management in the western US.

Evidence certainty: Moderate-to-high for total streamflow/water-yield increase (48 sources, multiple independent regions and methods); Low-to-moderate for baseflow-specific direction and magnitude in snow-dominated catchments specifically — directionally inconsistent findings persist even with the expanded evidence base, though Walvoord et al. (2025)’s synthesis provides a more structured account of *why* than was available previously. **Translation readiness:** Preliminary-to-intermediate — sufficient to motivate targeted instrumentation/study design and to identify the specific moderating variables (time since fire, burn severity, repeat-burn history, climate decoupling) most likely to explain site-to-site variation; not yet sufficient to support generalized predictive statements about baseflow direction at unstudied sites.

6. Declarations

Data Availability

Search log and candidate source list: `sources/search_log.md` (this repository). No code/data beyond search documentation was generated.

Methodology Deviation Note

This review substituted Claude Code’s native WebSearch/WebFetch tools for the `parallel-cli` tool specified as primary in the literature-review skill, which was not installed in this environment. PDF generation tooling (pandoc, xelatex) was available and used per the skill’s standard workflow.

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Review Metadata

Search dates: 2026-06-17, 2026-06-18 **Version:** 2.0 **Tooling deviation:** WebSearch/WebFetch substituted for parallel-cli (Round 1); **paper-search** CLI used directly per skill (Round 2)

Quality checks: - [x] Citations verified with verify_citations.py - [x] Search strategy documented (sources/search_log.md) - [x] Inclusion/exclusion criteria stated - [x] Thematic (not study-by-study) synthesis - [x] PDF generated